

WENYAN HU



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[Personal Web](#)

PERSONAL PROFILE

- **Senior Spatial Analyst** with a PhD in Geomatic Engineering, **three years of industry experience** in the Victorian Public Service, and **seven years of research experience** in spatial analysis, management, modelling, web development and system admin.
- Strong capability in ArcGIS Suite, Python, SQL, BIM, FME, QGIS, and cloud-based GIS environments, with extensive experience in supporting enterprise platforms, integrating complex spatial datasets, improving workflows, and delivering fit-for-purpose spatial solutions for multidisciplinary teams.

EXPERIENCE

Senior GIS Consultant, Digital Services

WSP | July 2026 – Current

Deliver GIS support for multidisciplinary infrastructure and environmental projects, with a focus on spatial data preparation, cartographic outputs, field data applications, and workflow improvement.

Key Tasks, Roles & Responsibilities:

- **Cartography and Project Delivery:** Produce high-quality **ecology and transport figures** for technical reports, environmental assessments, planning documentation, and client presentations, applying project-specific cartographic standards and quality review processes.
- **Field GIS Development:** Publish, configure, and enhance **ArcGIS Field Maps** and associated web maps to support field data collection, including layer configuration, form design, data access, publishing, and workflow improvements.
- **Spatial Development and Automation:** Develop **Python-based geoprocessing tools and scripts** to automate repetitive case-processing and data-handling tasks, improving consistency, reducing manual effort, and shortening project turnaround times.
- **Stakeholder and Project Collaboration:** Work closely with ecologists, transport specialists, engineers, and other project team members to understand requirements, respond to changing priorities, and deliver fit-for-purpose GIS products within consulting timeframes.

Senior Spatial Analyst, Spatial Service, Data and Digital

Department of Transport and Planning (DTP) | Jul 2023 – July 2026

Provided spatial services to support various state planning projects within DTP. Tasks include spatial data management, ETL automation, spatial analysis, cartographic mapping, web application development, and solution design to meet stakeholder needs for high-quality map-based products, reliable enterprise services, and data-driven insights.

Key Tasks, Roles & Responsibilities:

- **Data Management, Workflow Automation and Quality Assurance:** Built, transformed, validated, and integrated spatial datasets across enterprise systems, map services, and web applications, and developed Python-based ETL and data synchronisation workflows to improve data timeliness, reduce manual effort, and enhance the reliability of validation, publishing, and reporting processes.
- **ArcGIS Platform Support:** Supported AWS EC2-hosted ArcGIS Enterprise environments, including ArcGIS Server, Portal, Data Store, and Web Adaptor, ensuring reliable system performance and data availability, and used ArcGIS Monitor to monitor server health, review service performance, investigate issues, and support troubleshooting to maintain reliable system performance and data availability.
- **Spatial Analysis:** Led 2D and 3D spatial analysis for planning, transport, and environmental projects, including Public Transport Accessibility Level (PTAL) analysis, Definition Plan spatial analysis, LiDAR-based tree canopy extraction and analysis using a Tree Detection Deep Learning Model, and viewshed analysis for the Great Ocean Road, delivering decision-ready outputs for technical and non-technical stakeholders.
- **Spatial Development:** Led and contributed to the development, maintenance, and enhancement geoprocessing tools, add-ins, and planning applications to support statutory planning and map production workflows. This included Definition Plan, Planning Certificate, and Multi-Extrude GP tools developed in Python, as well as amendment and gazetted map printing add-ins developed in Python and C#.
- **Web Development:** Developed, maintained, and enhanced web GIS applications and map-based tools including [Vic3D](#), [Activity Centres Program](#), [Renewables Summary Page](#), [Public Transport Accessibility Level \(PTAL\)](#), Heritage, and [Community and Engagement](#) web maps. Responsibilities included map service publishing, user management, function design, and function enhancement, improving application usability, reliability, and access to spatial information for internal and public users.

- **Stakeholder Engagement:** Worked closely with planners, engineers, policy analysts, and data custodians to understand needs, refine requirements, and deliver fit-for-purpose data solutions.

Teaching Assistant, University of Melbourne

School of Electrical, Mechanical and Infrastructure Engineering | Dec 2019 – Apr 2024

Supported undergraduate and postgraduate teaching in spatial and engineering subjects, delivering practical instruction in QGIS, ArcGIS, spatial project design, simple flood simulation and technical report writing, while also co-supervising master's capstone projects and developing workshop materials.

Key Tasks, Roles & Responsibilities:

- Delivered **tutorials and practical support** for undergraduate and postgraduate subjects in Geographic Information Systems, Geospatial Informatics, and Engineering Modelling.
- Taught students to use **QGIS** and **ArcGIS** for spatial data processing, mapping, analysis and project work.
- **Designed** workshop handouts and teaching materials to support practical learning and improve student understanding of GIS concepts and workflows.
- Provided **troubleshooting** and **problem-solving** support for GIS software, spatial analysis tasks and technical issues.
- **Co-supervised** master's capstone students, providing technical guidance on geospatial analysis, project development and final deliverables.

EDUCATION

Doctor of Philosophy, Engineering – University of Melbourne, 2019-2023

Thesis: Information sensing, transmission and management between connected vehicles

Master of Science, Physical Geography – East China Normal University, 2016-2019

Visiting Scholar, GIS – Louisiana State University, 2018

Bachelor of Science, GIS – Northwest A&F University, 2012-2016

SKILLS

Programming:

- **Languages:** Python, JavaScript, TypeScript, C#, Java, SQL, Node.js

- **Libraries and Tools:** ArcGIS Maps SDK for JavaScript, ArcGIS API for Python, ArcGIS Maps SDK for Unreal Engine, Pandas, Numpy, Matplotlib, Seaborn, Plotly, PyTorch, TensorFlow, Geopandas, MATSim
- **Agile Tools:** Git, Jira, Confluence, Jenkins (CI/CD)

Software and Platforms:

- **ArcGIS Suite:** ArcMap, ArcGIS Pro, ArcGIS Online, ArcGIS Enterprise, ArcGIS Urban, ArcGIS Monitor, ArcGIS Dashboard, ArcGIS Survey123, ArcGIS Experience Builder
- **Other Spatial Tool:** QGIS, CloudCompare, FME, ENVI
- **Cloud Computing Services:** AWS
- **3D Modelling:** CityEngine, SketchUp, Rhino

PROFESSIONAL DEVELOPMENT

- **Spatial Analysis** with ArcGIS Pro (Esri)
- **Working With LiDAR Data** Using ArcGIS (Esri)
- **Imagery Analysis** Using ArcGIS Pro (Esri)
- **FME Form** Advanced Training (1 Spatial)
- **FME Flow** Authoring Training (1 Spatial)
- **Architecture on AWS** (Lumify)
- **Developing on AWS** (Lumify)
- **Developing Serverless Solutions on AWS** (Lumify)

PUBLICATIONS

- Luo, H., Hu, W., et al., 2025. Cross-sensor adaptive semantic segmentation for mobile laser scanning point clouds based on continuous potential scene surface reconstruction. *ISPRS Journal of Photogrammetry and Remote Sensing*.
- Chen, X., Hu, W., et al., 2025. Class-Incremental Semantic Segmentation for Mobile Laser Scanning Point Clouds Using Feature Representation Preservation and Loss Cross- Coupling. *Remote Sensing*.
- Hu, W., et al, 2023. Forecasting fine-grained sensing coverage in opportunistic vehicular sensing. *Computers, Environment and Urban Systems*.
- Hu, W., et al., 2024. Real-time taxi spatial anomaly detection based on vehicle trajectory prediction. *Travel Behaviour and Society*.
- Hu, W., et al., 2022. Decentralized management of ephemeral traffic incidents. *Transactions in GIS*.
- Hu, W., et al., 2022. Decentralized spreading of ephemeral road incident information between vehicles. *Journal of Intelligent Transportation Systems*.
- Hu W., et al., 2020. Impact of traffic on the spatiotemporal variations of spatial accessibility of emergency medical services in inner-city Shanghai. *Environment and Planning B: Urban Analytics and City Science*.
- Li, M., Hu, W., et al., 2023. Examining the effects of station-level factors on metro ridership using multiscale geographically weighted regression. *Journal of Transport Geography*.

AWARDS

- National Scholarship (2017) – awarded to the top 3% of students
- Third Prize, National Mathematical Contest in Modelling (2017) – project: UAV flight path design
- Recipient, ITSS 2022 Young Professionals Travelling Fellowship Program (2022)
- 2025 PIA AWARD WINNING - PLANNING WITH COUNTRY – project: PROGRESSIVE PLANNING FOR WATERWAYS

REFEREES

- Justin G Madex – Director GIS Professional Services Delivery, Woolpert
- Tim Gray – Lead Spatial Analyst, DTP